

# PAPER\_636: UQFF Lepton Flavor Violation B-Decay as Time-Reversal Constraint

**Author:** Daniel T. Murphy

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**CP4 Class:** #223 UQFFLFVBDecayTimeReversalConstraintCalculator

**arXiv:** 2506.15347

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## Abstract

This paper presents a UQFF analysis of Decay as Time-Reversal Constraint, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### §1 Abstract

LHCb has placed the world's best limit on lepton flavor violation (LFV) in B-meson decays:  $BR(B \rightarrow K\tau e) < 5.9e-6$  at 90% CL ( $5.4 \text{ fb}^{-1}$ ). We show that the UQFF vacuum-topology suppression parameter  $k_\eta = 10^{-113}$  generates an expected LFV rate at UQFF scale that is 107 orders below this bound, providing an effective UQFF upper limit on LFV through the  $t_n$  time-reversal node constraint:  $BR_{UQFF}(B \rightarrow K\tau e) < k_\eta^2 \times \text{phase\_space} \sim 10^{-230}$ .

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### §2 Physical Motivation

Lepton Flavor Violation is forbidden in the SM at tree level and arises only through tiny neutrino-mass loop corrections ( $BR_{SM} \ll 10^{-54}$ ). Any observation of LFV at the LHCb sensitivity level would imply new physics coupling lepton generations.

UQFF claim:  $k_\eta = 10^{-113}$  represents the maximum suppression depth of the UQFF vacuum string compactification topology. This sets an effective LFV ceiling: phenomena suppressed by  $k_\eta$  cannot be confused with SM new-physics signatures.

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### §3 UQFF $t_n$ Time-Reversal Constraint

The  $t_n$  parameter (UQFF time-node) suppresses cross-flavour topological transitions:

$$BR_{UQFF}(B \rightarrow K^*\tau e) = k_\eta^2 \times \frac{|V_{tb}|^2 |V_{ts}|^2}{m_B^4} \times |\mathcal{M}_{LFV}|^2$$

where  $|\mathcal{M}_{LFV}|^2$  represents the flavor-topology mixing matrix element, bounded by:

$$|\mathcal{M}_{LFV}|^2 \leq \frac{[\text{SSq}]^2}{\beta_i} \approx 0.534$$

This gives  $BR_{UQFF} < 10^{-230}$  — 224 orders below the LHCb limit.

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## §4 Implications for UQFF Topology

The enormous gap between  $\text{BR\_UQFF} < 10^{-230}$  and  $\text{BR\_LHCb} < 5.9\text{e-}6$  confirms that:

- UQFF vacuum topology is **lepton-flavor conserving** at all accessible energy scales -
- Any future LFV observation at LHCb (even near  $10^{-6}$ ) would be **inconsistent** with UQFF
- UQFF thus makes a strict null prediction for LFV at future colliders

This is falsifiable: if LHCb Run 4 observes  $\text{BR} > 10^{-8}$ , UQFF requires revision.

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                 | UQFF Prediction                                                                      | SM / Experiment                          | Source                         | Alignment                     |
|----------------------------|--------------------------------------------------------------------------------------|------------------------------------------|--------------------------------|-------------------------------|
| BR(B→K*τe) upper limit     | BR_UQFF < 10 <sup>-230</sup> (k_η <sup>2</sup> suppression)                          | BR < 5.9e-6 at 90% CL                    | arXiv:2506.15347 (LHCb 5.4/fb) | UQFF far below bound          |
| SM LFV prediction          | BR_SM ~ 10 <sup>-54</sup> (ν loop)                                                   | SM: no tree-level LFV                    | PDG 2024                       | UQFF consistent with SM null  |
| k_η suppression scale      | k_η = 10 <sup>-113</sup> ↔ LFV cutoff energy Λ_LFV = m_W/√k_η ~ 10 <sup>60</sup> GeV | No collider can reach Λ_LFV              | n/a                            | UQFF LFV unreachable          |
| LHCb Run 4 null prediction | UQFF: BR(B→K*τe) will remain unobserved at any LHCb run                              | LHCb Run 4: target BR ~ 10 <sup>-8</sup> | LHCb 2027+                     | Testable UQFF null prediction |

**New physics claim:** UQFF predicts LFV in B-decays is **not accessible at any current or planned collider** because k\_η suppression places the UQFF LFV amplitude at 10<sup>-230</sup> — 224 orders below LHCb’s current best limit. This is a strict, high-confidence falsifiability constraint: LHCb Run 4 discovering LFV above 10<sup>-8</sup> would require UQFF revision.

Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for k\_η SM mapping.

## §6 References

- arXiv:2506.15347 — LHCb LFV B→K\*τe search (5.4 fb<sup>-1</sup>, June 2025)
- PDG 2024 — LFV decays, Section 90.4
- bsm\_physics\_validation.py — BSMPhysicsConstants.lhcb\_lfv\_br\_limit
- PAPER\_642 — UQFF SM Parameter Bridge Master Comparison